Semester	B.E. Semester VII – Computer Engineering
Subject	DevOps Lab
Subject In-charge	Dr. Snehal Andhare

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Task Number	6
Project Title:	Secure CI/CD Pipeline for a Loan Application Tracking Portal
Task Title	MVP Completion and Git Collaboration: Implement the remaining essential MVP functions for the Loan Application Tracking Portal: data/event entry, searchable dashboard, summary indicators, status drill-down and alert/exception view. Demonstrate a second feature branch, conflict creation/resolution, tagging and a release-ready source baseline.

Deliverable:	**All the remaining functional MVP features—Summary KPI Indicators, Searchable Dashboard, Status Drill-Down with Event Timeline, and the Alerts/Exceptions View:**
	1. `details.html` - **Status Drill-Down View** showing borrower profile, timeline of events, and stage transition controls.
	2. `alerts.html` - **Alerts & Exception View** displaying rejected loans with reasons.
	3. `dashboard.html` - Upgraded with **4 KPI Summary Indicator Cards**, interactive search bar, and action links.
	4. `LoanController.java` - Extended with routes for status drill-down, stage updates, and alerts.
	5. `LoanServiceTest.java` - Comprehensive test suite covering search, status updates, and audit event logging.
	
	1. Create Second Feature Branch & Commit MVP Code:
	
	`D:\Vicky_Project\DevOps\DevOps_Project>git checkout develop`
	`M .gitignore`
	`M src/main/java/com/loanportal/controller/LoanController.java`
	`M src/main/resources/application.properties`
	`M src/main/resources/templates/dashboard.html`
	`M src/test/java/com/loanportal/service/LoanServiceTest.java`
	`Already on 'develop'`
	`Your branch is up to date with 'origin/develop'.`

```
D:\Vicky_Project\DevOps\DevOps_Project>git checkout -b feature/analytics-and-alerts
Switched to a new branch 'feature/analytics-and-alerts'

D:\Vicky_Project\DevOps\DevOps_Project>git add .
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/loanportal/controller/LoanController.java', LF will be replaced by CR
LF the next time Git touches it
warning: in the working copy of 'src/main/resources/application.properties', LF will be replaced by CRLF the next time G
it touches it
warning: in the working copy of 'src/main/resources/templates/dashboard.html', LF will be replaced by CRLF the next time
 Git touches it
warning: in the working copy of 'src/test/java/com/loanportal/service/LoanServiceTest.java', LF will be replaced by CRLF
 the next time Git touches it
warning: in the working copy of 'src/main/resources/templates/alerts.html', LF will be replaced by CRLF the next time Gi
t touches it
warning: in the working copy of 'src/main/resources/templates/details.html', LF will be replaced by CRLF the next time G
it touches it

D:\Vicky_Project\DevOps\DevOps_Project>git commit -m "feat(mvp): complete dashboard indicators, status drill-down, and a
lerts view"
[feature/analytics-and-alerts 2001a59] feat(mvp): complete dashboard indicators, status drill-down, and alerts view
 9 files changed, 386 insertions(+), 24 deletions(-)
 create mode 100644 DOC/DevOps_Task5_Vicky_Pukale_23102A0033.docx
 create mode 100644 PDF/DevOps_Task5_Vicky_Pukale_23102A0033.pdf
 create mode 100644 src/main/resources/templates/alerts.html
 create mode 100644 src/main/resources/templates/details.html

D:\Vicky_Project\DevOps\DevOps_Project>git push -u origin feature/analytics-and-alerts
Enumerating objects: 46, done.
Counting objects: 100% (46/46), done.
Delta compression using up to 16 threads
Compressing objects: 100% (19/19), done.
Writing objects: 100% (26/26), 1.47 MiB | 2.10 MiB/s, done.
Total 26 (delta 7), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (7/7), completed with 7 local objects.
remote:
remote: Create a pull request for 'feature/analytics-and-alerts' on GitHub by visiting:
remote:      https://github.com/vickypukale/loan-application-tracking-portal/pull/new/feature/analytics-and-alerts
remote:
To https://github.com/vickypukale/loan-application-tracking-portal.git
 * [new branch]      feature/analytics-and-alerts -> feature/analytics-and-alerts
branch 'feature/analytics-and-alerts' set up to track 'origin/feature/analytics-and-alerts'.
```

2. Create a Deliberate Merge Conflict (For Project Demonstration):

- **Switch to `develop` and make a competing change to `README.md`:**

```
D:\Vicky_Project\DevOps\DevOps_Project>git checkout develop
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
```

Add a status line to `README.md` by running:

```
D:\Vicky_Project\DevOps\DevOps_Project>git checkout develop
Already on 'develop'
Your branch is up to date with 'origin/develop'.

D:\Vicky_Project\DevOps\DevOps_Project>echo ### Status: Under Active Review in Sprint 2 >> README.md

D:\Vicky_Project\DevOps\DevOps_Project>git add README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it

D:\Vicky_Project\DevOps\DevOps_Project>git commit -m "docs: update development status badge on develop branch"
[develop 6b594b1] docs: update development status badge on develop branch
 1 file changed, 1 insertion(+)
```

Switch back to feature/analytics-and-alerts and make a conflicting change to the same line:

```
D:\Vicky_Project\DevOps\DevOps_Project>git checkout feature/analytics-and-alerts
Switched to branch 'feature/analytics-and-alerts'
Your branch is up to date with 'origin/feature/analytics-and-alerts'.

D:\Vicky_Project\DevOps\DevOps_Project>echo ### Status: Release-Ready Baseline v1.0.0 >> README.md

D:\Vicky_Project\DevOps\DevOps_Project>git add README.md

D:\Vicky_Project\DevOps\DevOps_Project>git commit -m "docs: update production release status on feature branch"
[feature/analytics-and-alerts 1a8760b] docs: update production release status on feature branch
 1 file changed, 1 insertion(+)
```

3. Trigger the merge conflict:

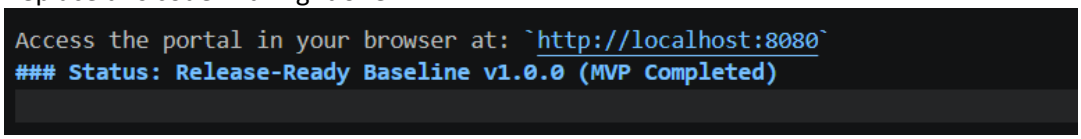
```
D:\Vicky_Project\DevOps\DevOps_Project>git checkout develop
Switched to branch 'develop'
Your branch is ahead of 'origin/develop' by 1 commit.
  (use "git push" to publish your local commits)

D:\Vicky_Project\DevOps\DevOps_Project>git merge feature/analytics-and-alerts
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.
```

conflict markers in Editor(like VS code):



Replace this code with right one :



Now finalize the merge in terminal:

```
D:\Vicky_Project\DevOps\DevOps_Project>git add .

D:\Vicky_Project\DevOps\DevOps_Project>git commit -m "fix(merge): resolve conflict between develop and feature/analytics
-and-alerts"
[develop 3155ed1] fix(merge): resolve conflict between develop and feature/analytics-and-alerts

D:\Vicky_Project\DevOps\DevOps_Project>git push origin develop
Enumerating objects: 13, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 16 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (9/9), 1.00 KiB | 1.00 MiB/s, done.
Total 9 (delta 6), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (6/6), completed with 3 local objects.
To https://github.com/vickypukale/loan-application-tracking-portal.git
   4b501a6..3155ed1  develop -> develop
```

4. Merge into `main` and Tag the Version (`v1.0.0`):

```
D:\Vicky_Project\DevOps\DevOps_Project>git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

D:\Vicky_Project\DevOps\DevOps_Project>git merge develop
Updating d9c66ab..3155ed1
Fast-forward
 .gitignore                                    |   1 +
 DOC/DevOps_Task4_Vicky_Pukale_23102A0033.docx | Bin 0 -> 490920 bytes
 DOC/DevOps_Task5_Vicky_Pukale_23102A0033.docx | Bin 0 -> 852752 bytes
 PDF/DevOps_Task4_Vicky_Pukale_23102A0033.pdf | Bin 0 -> 631979 bytes
 PDF/DevOps_Task5_Vicky_Pukale_23102A0033.pdf | Bin 0 -> 760882 bytes
 README.md                                     |   1 +
 .../com/loanportal/controller/LoanController.java  |  71 +++++++++
 .../java/com/loanportal/model/LoanApplication.java | 100 ++++++++++++
 src/main/java/com/loanportal/model/LoanEvent.java  |  51 +++++++
 .../repository/LoanApplicationRepository.java      |  16 ++
 .../loanportal/repository/LoanEventRepository.java |  13 ++
 .../java/com/loanportal/service/LoanService.java   |  99 ++++++++++++
 src/main/resources/application.properties          |   2 +-
 src/main/resources/templates/alerts.html           |  77 ++++++++++
 src/main/resources/templates/apply.html            |  74 ++++++++++
 src/main/resources/templates/dashboard.html        | 132 ++++++++++++++++
 src/main/resources/templates/details.html          | 162 ++++++++++++++++++++
+++
 .../com/loanportal/service/LoanServiceTest.java    |  97 ++++++++++++
 18 files changed, 895 insertions(+), 1 deletion(-)
 create mode 100644 DOC/DevOps_Task4_Vicky_Pukale_23102A0033.docx
 create mode 100644 DOC/DevOps_Task5_Vicky_Pukale_23102A0033.docx
 create mode 100644 PDF/DevOps_Task4_Vicky_Pukale_23102A0033.pdf
 create mode 100644 PDF/DevOps_Task5_Vicky_Pukale_23102A0033.pdf
 create mode 100644 src/main/java/com/loanportal/controller/LoanController.j
ava
 create mode 100644 src/main/java/com/loanportal/model/LoanApplication.java
 create mode 100644 src/main/java/com/loanportal/model/LoanEvent.java
 create mode 100644 src/main/java/com/loanportal/repository/LoanApplicationR
epository.java
 create mode 100644 src/main/java/com/loanportal/repository/LoanEventReposit
ory.java
 create mode 100644 src/main/java/com/loanportal/service/LoanService.java
 create mode 100644 src/main/resources/templates/alerts.html
 create mode 100644 src/main/resources/templates/apply.html
 create mode 100644 src/main/resources/templates/dashboard.html
 create mode 100644 src/main/resources/templates/details.html
 create mode 100644 src/test/java/com/loanportal/service/LoanServiceTest.jav
```

```
D:\Vicky_Project\DevOps\DevOps_Project>git tag -a v1.0.0 -m "Release v1.0.0:
 Function MVP of Loan Applicaation Tracking Portal"

D:\Vicky_Project\DevOps\DevOps_Project>git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/vickypukale/loan-application-tracking-portal.git
   d9c66ab..3155ed1  main -> main

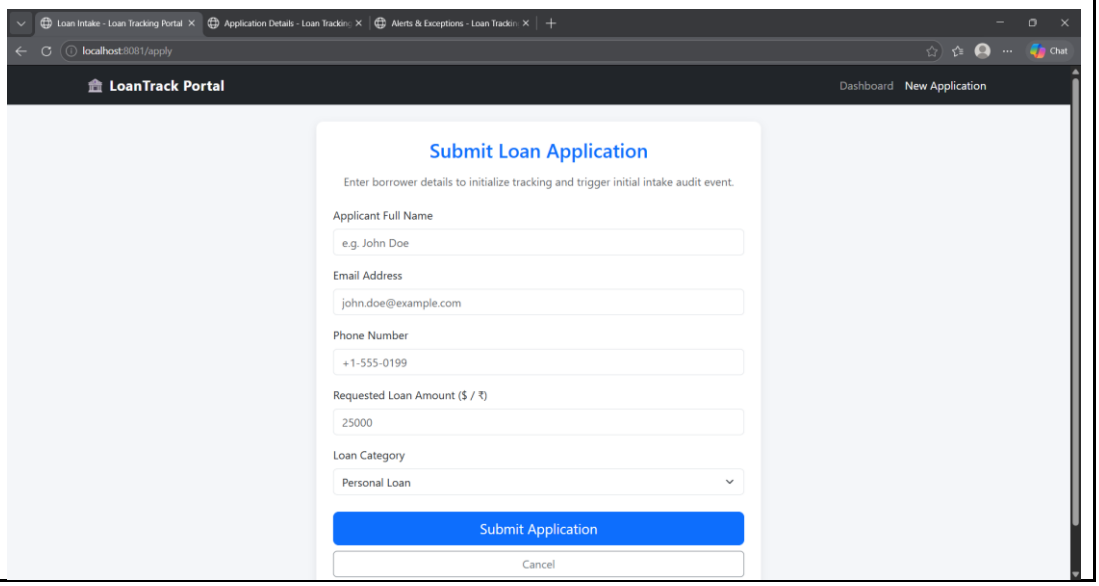
D:\Vicky_Project\DevOps\DevOps_Project>git push origin v1.0.0
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 216 bytes | 216.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/vickypukale/loan-application-tracking-portal.git
 * [new tag]         v1.0.0 -> v1.0.0

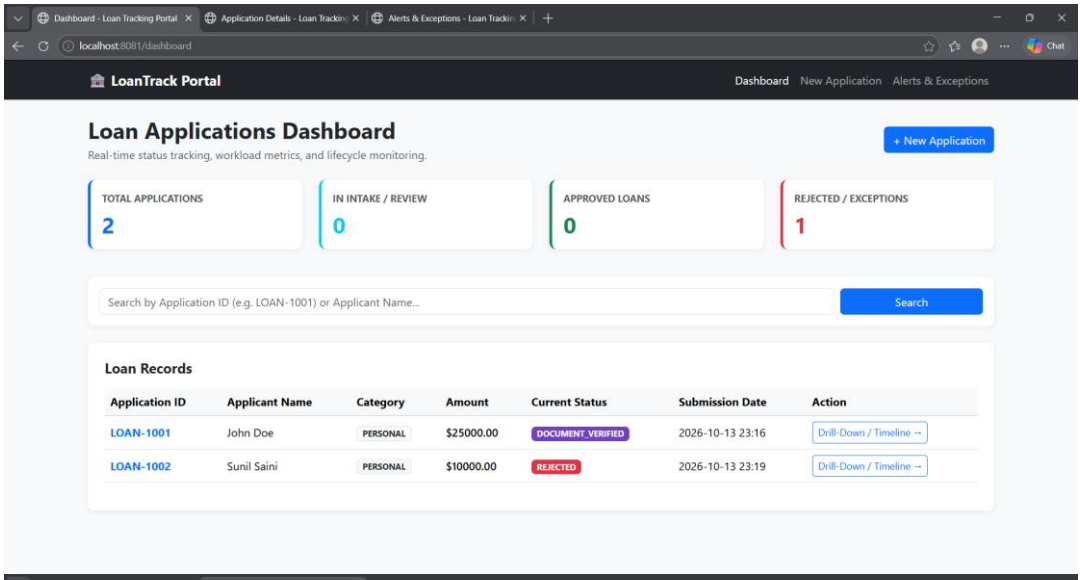
D:\Vicky_Project\DevOps\DevOps_Project>git tag -n
v1.0.0          Release v1.0.0: Function MVP of Loan Applicaation Tracking P
ortal
```

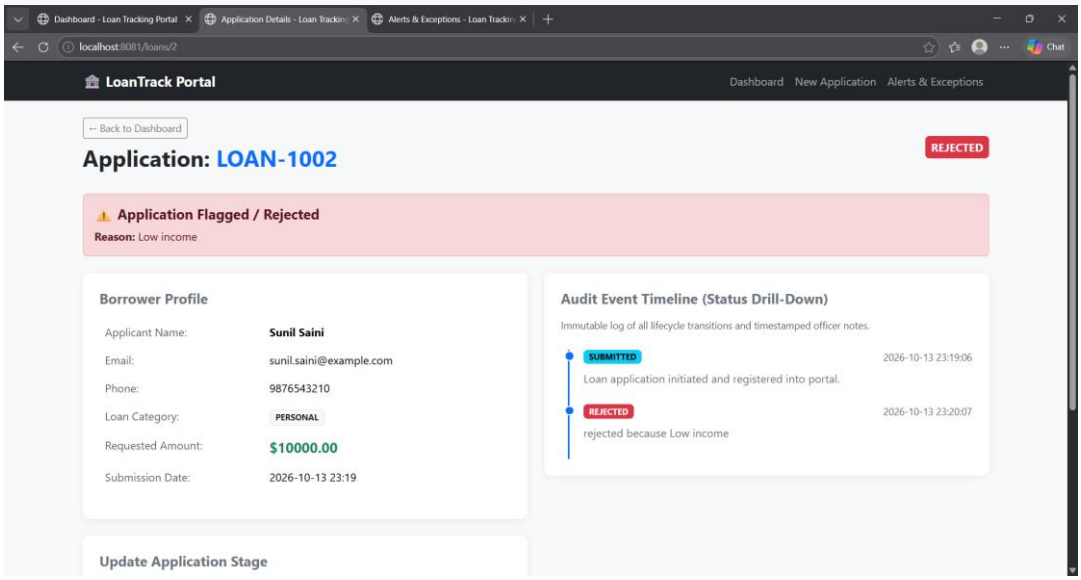
5. Verify the MVP Locally:

Run: mvn clean run

```
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 6.248
 s -- in com.loanportal.LoanApplicationPortalAppTests
[INFO] Running com.loanportal.service.LoanServiceTest
2026-10-13T23:14:24.939+05:30  INFO 21440 --- [LoanApplicationTrackingPortal
] [           main] t.c.s.AnnotationConfigContextLoaderUtils : Could not det
ect default configuration classes for test class [com.loanportal.service.Loa
nServiceTest]: LoanServiceTest does not declare any static, non-private, non
-final, nested classes annotated with @Configuration.
2026-10-13T23:14:24.947+05:30  INFO 21440 --- [LoanApplicationTrackingPortal
] [           main] .b.t.c.SpringBootTestContextBootstrapper : Found @Spring
BootConfiguration com.loanportal.LoanApplicationPortalApp for test class com
.loanportal.service.LoanServiceTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.532
 s -- in com.loanportal.service.LoanServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
---
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
---
[INFO] Total time:  11.429 s
[INFO] Finished at: 2026-10-13T23:14:25+05:30
[INFO] ------------------------------------------------------------------------
---
```




Dashboard - Loan Tracking Portal
Application Details - Loan Trackin
Alerts & Exceptions - Loan Trackin
localhost:8081/dashboard
LoanTrack Portal
Dashboard
New Application
Alerts & Exceptions
Loan Applications Dashboard
Real-time status tracking, workload metrics, and lifecycle monitoring.
+ New Application
TOTAL APPLICATIONS
2
IN INTAKE / REVIEW
0
APPROVED LOANS
0
REJECTED / EXCEPTIONS
1
Search by Application ID (e.g. LOAN-1001) or Applicant Name...
Search
Loan Records
Application ID
Applicant Name
Category
Amount
Current Status
Submission Date
Action
LOAN-1001
John Doe
PERSONAL
$25000.00
DOCUMENT_VERIFIED
2026-10-13 23:16
Drill-Down / Timeline →
LOAN-1002
Sunil Saini
PERSONAL
$10000.00
REJECTED
2026-10-13 23:19
Drill-Down / Timeline →


Dashboard - Loan Tracking Portal
Application Details - Loan Trackin
Alerts & Exceptions - Loan Trackin
localhost:8081/loans/2
LoanTrack Portal
Dashboard
New Application
Alerts & Exceptions
← Back to Dashboard
Application: LOAN-1002
REJECTED
Application Flagged / Rejected
Reason: Low income
Borrower Profile
Applicant Name:
Sunil Saini
Email:
sunil.saini@example.com
Phone:
9876543210
Loan Category:
PERSONAL
Requested Amount:
$10000.00
Submission Date:
2026-10-13 23:19
Audit Event Timeline (Status Drill-Down)
Immutable log of all lifecycle transitions and timestamped officer notes.
SUBMITTED
2026-10-13 23:19:06
Loan application initiated and registered into portal.
REJECTED
2026-10-13 23:20:07
rejected because Low income
Update Application Stage